Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Frank Chaves and Sons #2

Physical address of dairy:

Number and Street	City	County	Zip Code
12021 E Harney LN	Lodi	San Joaquin	95240

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/05/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X063-X250-XX16-XXXX

B. OPERATORS

Chaves, Laurinda

Operator name: Chaves, Laurinda

Telephone no.: (209) 368-1740 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12754 E Harney LN	Lodi	CA	95240

This operator is responsible for paying permit fees.

C. OWNERS

Chaves, Laurinda

Legal owner name: Chaves, Laurinda

Telephone no.: (209) 368-1740 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12754 E Harney LN	Lodi	CA	95240

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	20	0	50	0
Number under roof	200	0	0	0	0	0
Maximum number	200	0	20	0	50	0
Average number	200	0	20	0	50	0
Avg live weight (lbs)	1,300	0	1,000	0		

Predominant milk cow breed: Holstein

Average milk production: 51 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 5,064.05 *tons per reporting period*

Total nitrogen from manure: 65,698.07 *lbs per reporting period*

After ammonia losses (30% loss applied): 45,988.65 *lbs per reporting period*

Total phosphorus from manure: 10,785.29 *lbs per reporting period*

Total potassium from manure: 35,805.25 *lbs per reporting period*

Total salt from manure: 94,170.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 6,171,000 *gallons*

Total nitrogen generated: 15,667.19 *lbs*

Total phosphorus generated: 2,233.27 *lbs*

Total potassium generated: 19,320.74 *lbs*

Total salt generated: 119,694.34 *lbs*

	6,171,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	6,171,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
C2D1	Ground water
C2I1	Ground water
CR1	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
C21	40	40	2	both	X630-X250-XX16-XXXX
CR70	70	70	4	both	X630-X250-XX01-XXXX
Totals for areas that were used for application	110	110	6		
Totals for areas that were not used for application					
Land application area totals	110	110	6		

B. CROPS AND HARVESTS

C21

Field name: C21

10/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 40 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	580.00 *ton*	Dry-weight		72.6	11,200.00	3,200.00	45,000.00		9.66

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	91.50	27.00	184.50	0.00
Total actual harvest content	14.50	89.00	25.43	357.57	767.58

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	1,120.00 *ton*	Dry-weight		68.8	12,000.00	2,640.00	16,600.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	195.00	41.60	267.80	0.00
Total actual harvest content	28.00	209.66	46.13	290.04	1,520.06

CR70

CR70

Field name: CR70

10/27/2022: Winter Forage

Crop: Winter Forage Acres planted: 70 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	580.00 *ton*	Dry-weight		76.6	14,600.00	3,400.00	50,000.00		7.29
05/09/2023	435.00 *ton*	Dry-weight		76.4	14,500.00	3,000.00	43,000.00		9.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	102.00	24.00	175.50	0.00
Total actual harvest content	14.50	99.15	21.98	320.01	572.77

06/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 70 Plant date: 06/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	1,280.00 *ton*	Dry-weight		73.2	12,400.00	2,510.00	15,000.00		8.80
10/13/2023	930.00 *ton*	Dry-weight		72.8	13,400.00	2,530.00	12,300.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	189.80	36.40	174.20	0.00
Total actual harvest content	31.57	218.38	42.89	235.91	1,512.97

NUTRIENT BUDGET

A. LAND APPLICATIONS

C21 - 10/28/2022: Winter Forage

Field name: C21

Crop: Winter Forage — Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	94.87	34.98	267.84	1,116.33	120.00 *ton*
Application event totals		94.87	34.98	267.84	1,116.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/27/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	19.36	6.07	35.13	163.22	663,000.00 *gal*
Application event totals		19.36	6.07	35.13	163.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	19.36	6.07	35.13	163.22	663,000.00 *gal*
Application event totals		19.36	6.07	35.13	163.22	

C21 - 06/14/2023: Corn, silage

Field name: C21

Crop: Corn, silage — Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C2I1	Ground water	1.90	0.00	0.00	0.00	4,788,000.00 *gal*
Application event totals		1.90	0.00	0.00	0.00	

C21 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	93.66	20.50	95.76	0.00	400.00 *ton*
Application event totals		93.66	20.50	95.76	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C2I1	Ground water	1.77	0.00	0.00	0.00	4,464,000.00 *gal*
Application event totals		1.77	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	51.25	3.61	50.65	374.52	408,000.00 *gal*
C2I1	Ground water	1.77	0.00	0.00	0.00	4,464,000.00 *gal*
Application event totals		53.02	3.61	50.65	374.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C2I1	Ground water	1.77	0.00	0.00	0.00	4,464,000.00 *gal*
Application event totals		1.77	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	51.25	3.61	50.65	374.52	408,000.00 *gal*
C2I1	Ground water	1.77	0.00	0.00	0.00	4,464,000.00 *gal*
Application event totals		53.02	3.61	50.65	374.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C2I1	Ground water	1.77	0.00	0.00	0.00	4,464,000.00 *gal*
Application event totals		1.77	0.00	0.00	0.00	

CR70 - 10/27/2022: Winter Forage

Field name: CR70

Crop: Winter Forage

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	94.87	34.98	267.84	1,116.33	210.00 *ton*
Application event totals		94.87	34.98	267.84	1,116.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/25/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	22.13	6.94	40.15	186.53	1,326,000.00 *gal*
Application event totals		22.13	6.94	40.15	186.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Pipeline	No precipitation	No precipitation	Steady rain

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	22.13	6.94	40.15	186.53	1,326,000.00 *gal*
Application event totals		22.13	6.94	40.15	186.53	

CR70 - 06/21/2023: Corn, silage

Field name: CR70

Crop: Corn, silage

Plant date: 06/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	93.66	20.50	95.76	0.00	700.00 *ton*
Application event totals		93.66	20.50	95.76	0.00	

CR70 - 06/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CR1	Ground water	3.94	0.00	0.00	0.00	7,878,000.00 *gal*
Application event totals		3.94	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CR1	Ground water	3.75	0.00	0.00	0.00	7,488,000.00 *gal*
Application event totals		3.75	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CR1	Ground water	3.75	0.00	0.00	0.00	7,488,000.00 *gal*
Application event totals		3.75	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	49.42	3.48	48.84	361.15	688,500.00 *gal*
CR1	Ground water	3.75	0.00	0.00	0.00	7,488,000.00 *gal*
Application event totals		53.17	3.48	48.84	361.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CR1	Ground water	3.75	0.00	0.00	0.00	7,488,000.00 *gal*
Application event totals		3.75	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	49.42	3.48	48.84	361.15	688,500.00 *gal*
CR1	Ground water	3.75	0.00	0.00	0.00	7,488,000.00 *gal*
Application event totals		53.17	3.48	48.84	361.15	

CR70 - 06/21/2023: Corn, silage

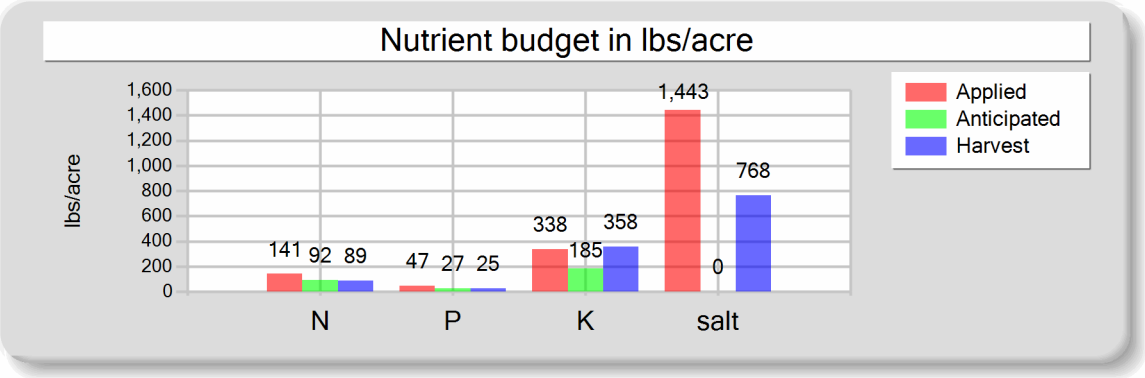
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CR1	Ground water	3.75	0.00	0.00	0.00	7,488,000.00 *gal*
Application event totals		3.75	0.00	0.00	0.00	

B. NUTRIENT BUDGET

C21 - 10/28/2022: Winter Forage

Field name: C21 Crop: Winter Forage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	94.87	34.98	267.84	1,116.33
Process wastewater	38.73	12.14	70.27	326.43
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	140.60	47.12	338.11	1,442.76
Anticipated crop nutrient removal	91.50	27.00	184.50	0.00
Actual crop nutrient removal	89.00	25.43	357.57	767.58
Nutrient balance	51.60	21.69	-19.46	675.18
Applied to removed ratio	1.58	1.85	0.95	1.88

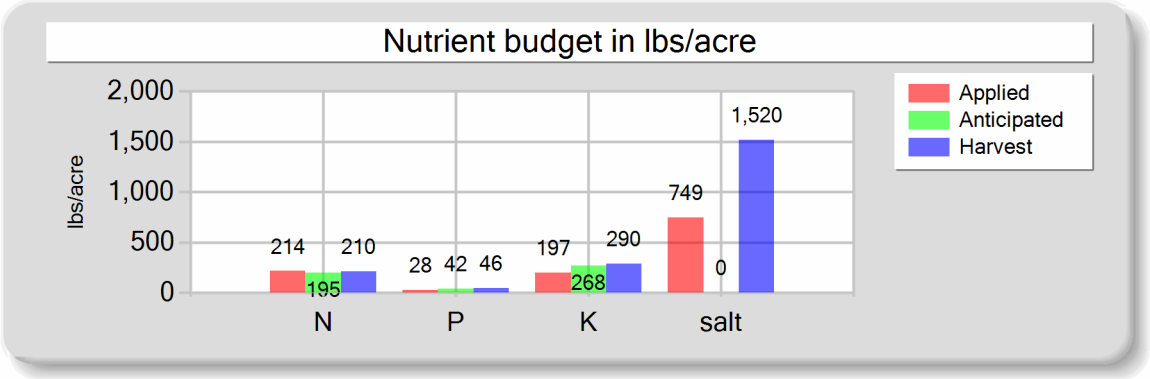
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,326,000.00 *gallons*
48.83 *acre-inches*
1.22 *inches/acre*

Total harvests for the crop
1 *harvests*

C21 - 06/14/2023: Corn, silage

Field name: C21 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	93.66	20.50	95.76	0.00
Process wastewater	102.51	7.22	101.29	749.05
Fresh water	10.75	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	213.91	27.71	197.05	749.05
Anticipated crop nutrient removal	195.00	41.60	267.80	0.00
Actual crop nutrient removal	209.66	46.13	290.04	1,520.06
Nutrient balance	4.25	-18.41	-92.98	-771.02
Applied to removed ratio	1.02	0.60	0.68	0.49

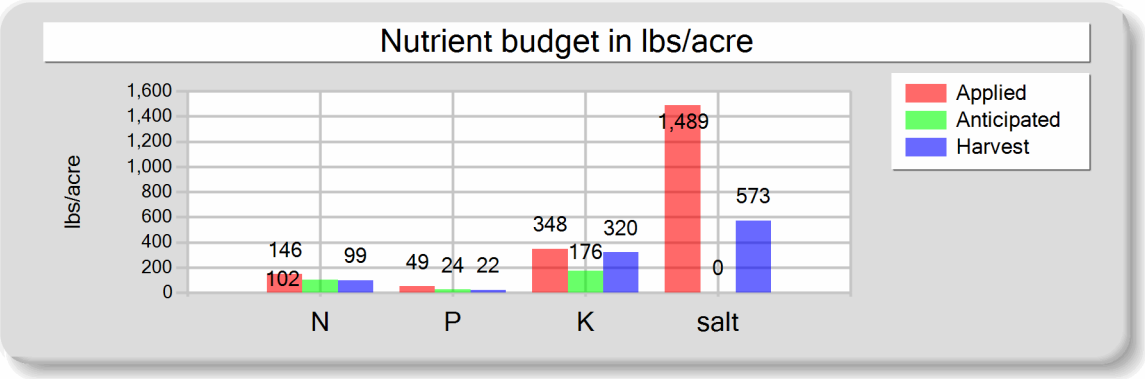
Fresh water applied	
27,108,000.00	*gallons*
998.30	*acre-inches*
24.96	*inches/acre*

Process wastewater applied	
816,000.00	*gallons*
30.05	*acre-inches*
0.75	*inches/acre*

Total harvests for the crop	
1	*harvests*

CR70 - 10/27/2022: Winter Forage

Field name: CR70 Crop: Winter Forage Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	94.87	34.98	267.84	1,116.33
Process wastewater	44.26	13.88	80.30	373.06
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	146.13	48.86	348.15	1,489.39
Anticipated crop nutrient removal	102.00	24.00	175.50	0.00
Actual crop nutrient removal	99.15	21.98	320.01	572.77
Nutrient balance	46.99	26.87	28.14	916.62
Applied to removed ratio	1.47	2.22	1.09	2.60

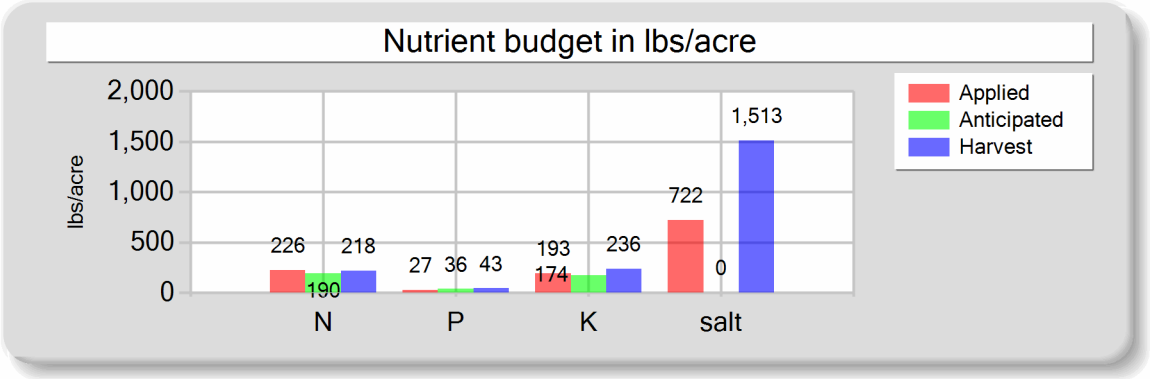
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
2,652,000.00 *gallons*
97.66 *acre-inches*
1.40 *inches/acre*

Total harvests for the crop
2 *harvests*

CR70 - 06/21/2023: Corn, silage

Field name: CR70 Crop: Corn, silage Plant date: 06/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	93.66	20.50	95.76	0.00
Process wastewater	98.85	6.96	97.67	722.30
Fresh water	26.44	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.95	27.46	193.43	722.30
Anticipated crop nutrient removal	189.80	36.40	174.20	0.00
Actual crop nutrient removal	218.38	42.89	235.91	1,512.97
Nutrient balance	7.57	-15.43	-42.48	-790.67
Applied to removed ratio	1.03	0.64	0.82	0.48

Fresh water applied	
52,806,000.00	*gallons*
1,944.67	*acre-inches*
27.78	*inches/acre*

Process wastewater applied	
1,377,000.00	*gallons*
50.71	*acre-inches*
0.72	*inches/acre*

Total harvests for the crop	
2	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2313020A

Sample and source description: T2313020A

Sample date: 11/09/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 36.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,900.00	9,180.00	70,300.00	10,300.00	8,330.00	4,060.00	6,600.00	818.00		29.30
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3132036D

Sample and source description: T3132036D

Sample date: 05/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,300.00	4,880.00	22,800.00							0.00
DL	1.00	1.00	1.00							1.00

T3264047B

Sample and source description: T3264047B

Sample date: 09/21/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 30.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,700.00	5,960.00	23,200.00	12,100.00	6,670.00	2,500.00	7,510.00	837.00		33.40
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3053011A

Sample and source description: T3053011A

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	140.00	86.80	0.00	0.00	43.90	254.00								2,060.00	1,180
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3132036E

Sample and source description: T3132036E

Sample date: 05/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	319.00	252.00	0.00	0.00	98.80	541.00								4,380.00	2,300
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3264047A

Sample and source description: T3264047A

Sample date: 09/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	602.00	434.00	0.00	0.15	42.40	595.00	122.00	78.30	146.00	3,000.00	0.00	171.00	262.00	6,660.00	4,400
DL	1.00	1.00	1.00	0.10	1.00	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

C. FRESH WATER ANALYSES

C2D1

C2D1

T3314017A

Sample description: T3314017A

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.09	0.00	2.09								264.00	0
DL	1.00	0.50	1.00								100.00	1

C2I1

T3314017B

Sample description: T3314017B

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.90	0.00	1.90								272.00	0
DL	1.00	0.50	1.00								100.00	1

CR1

T3314017C

Sample description: T3314017C

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.20	0.00	4.20								335.00	0
DL	1.00	0.50	1.00								100.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

C21 - 10/28/2022: Winter Forage

T3132036C

Sample and source description: T3132036C

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	3,200.00	45,000.00		9.66
DL	1.00	1.00	1.00		1.00

C21 - 06/14/2023: Corn, silage

T3293030A

Sample and source description: T3293030A

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	2,640.00	16,600.00		8.70
DL	1.00	1.00	1.00		1.00

CR70 - 10/27/2022: Winter Forage

CR70 - 10/27/2022: Winter Forage

T3132036A

Sample and source description: T3132036A

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	3,400.00	50,000.00		7.29
DL	1.00	1.00	1.00		1.00

T3132036B

Sample and source description: T3132036B

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,500.00	3,000.00	43,000.00		9.89
DL	1.00	1.00	1.00		1.00

CR70 - 06/21/2023: Corn, silage

T3293030B

Sample and source description: T3293030B

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,510.00	15,000.00		8.80
DL	1.00	1.00	1.00		1.00

CR70 - 06/21/2023: Corn, silage

T3293030C

Sample and source description: T3293030C

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	2,530.00	12,300.00		9.00
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

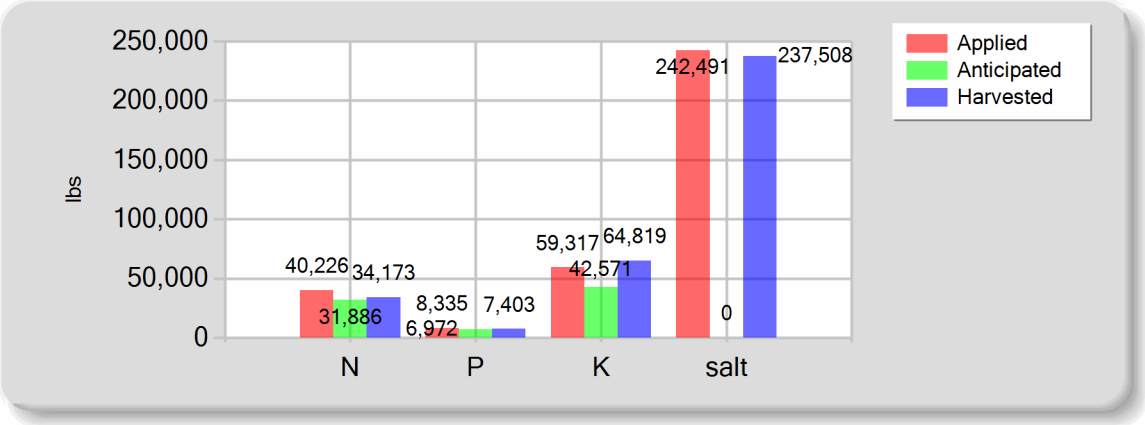
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

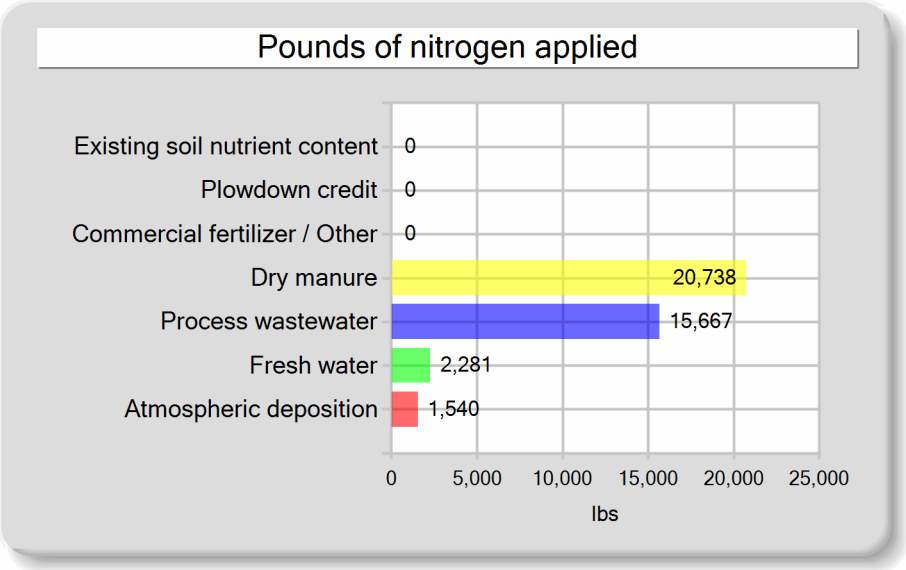
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

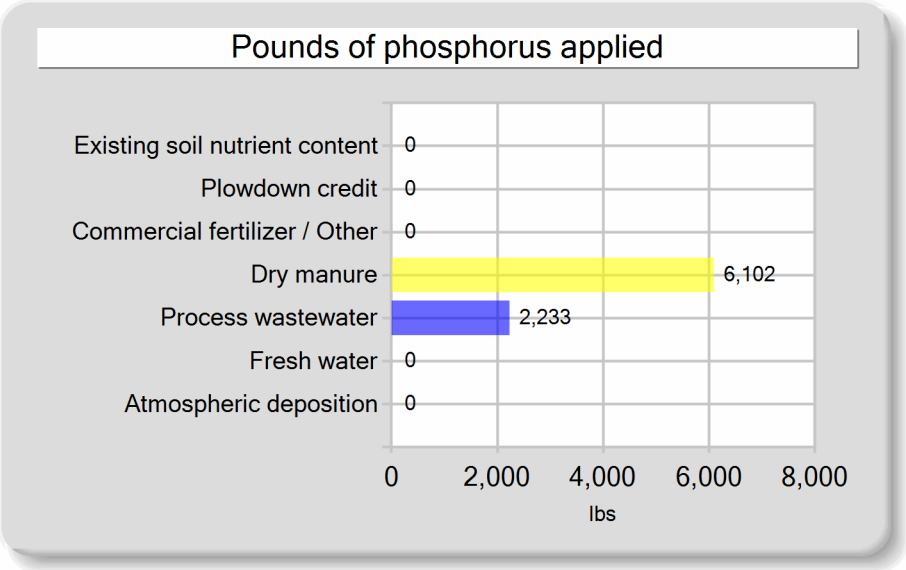
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	20,738.19	6,101.90	39,996.33	122,796.30
Process wastewater	15,667.19	2,233.27	19,320.74	119,694.34
Fresh water	2,280.61	0.00	0.00	0.00
Atmospheric deposition	1,540.00	0.00	0.00	0.00
Total nutrients applied	40,225.99	8,335.17	59,317.07	242,490.64
Anticipated crop nutrient removal	31,886.00	6,972.00	42,571.00	0.00
Actual crop nutrient removal	34,173.25	7,403.03	64,818.98	237,507.87
Nutrient balance	6,052.74	932.14	-5,501.92	4,982.77
Applied to removed ratio	1.18	1.13	0.92	1.02

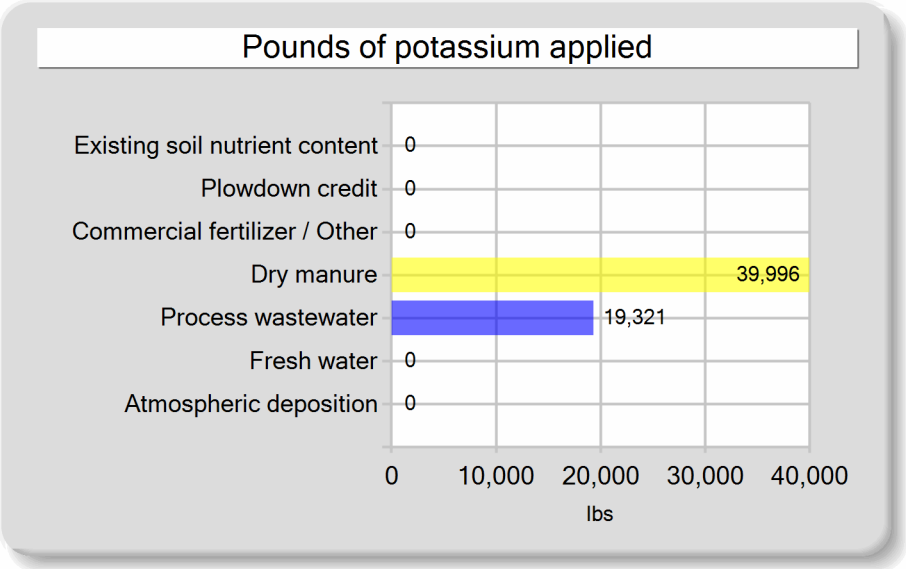
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

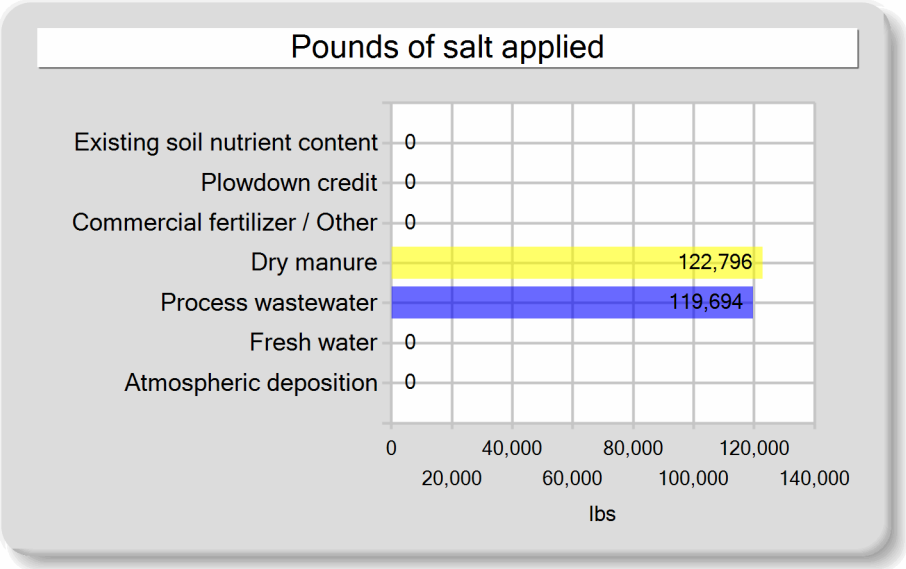


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Both C21 and CR70 had low plant tissue sample tests for winter forage. Applications were well below average applications with corn nutrient applications low as well. Excess nutrients in fields are used by the corn crop.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Laurinda S. Chaves	Laurinda S. Chaves
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Laurinda Chaves	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/19/24	6/19/24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: FRANK CHAVES & SONS DAIRY #2
Mailing address: 12754 E. HARNEY LANE
(not facility address) LODI, CA 95240
Phone#: (209) 329-4243

Authorized Copy Released To: JCRK, INC.
(209)327-0992
Sampled By: Stanley Chaves
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/9/23	10:00a	C2D1	Dom. Well	10.6°C	20min	X				
2	✓	10:30a	C2I1	Ag Well	10.4°	✓	X				
3	✓	11:00a	CR1	✓	10.2	✓	X				
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Jacquie Captein	JCRK, INC.		11/10/23 8:00a		
2nd		DENELE ANALYTICAL, INC.	11-10-23 1100			T3314017

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Frank Chaves & Sons Dairy #2
12754 E. Harney Ln.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 10.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
C2D1	T3314017A	Electrical Conductivity (EC)	EPA 120.1 and	0.264	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.09	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	169	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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40 N. East Street, Woodland, CA 95776
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Compliance Analysis Report

Customer
Frank Chaves & Sons Dairy #2
12754 E. Harney Ln.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 10.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
C2I1	T3314017B	Electrical Conductivity (EC)	EPA 120.1 and	0.272	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.90	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	174	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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Denele Analytical, Inc.

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Frank Chaves & Sons Dairy #2
12754 E. Harney Ln.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 10.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
CR1	T3314017C	Electrical Conductivity (EC)	EPA 120.1 and	0.335	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.20	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	214	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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